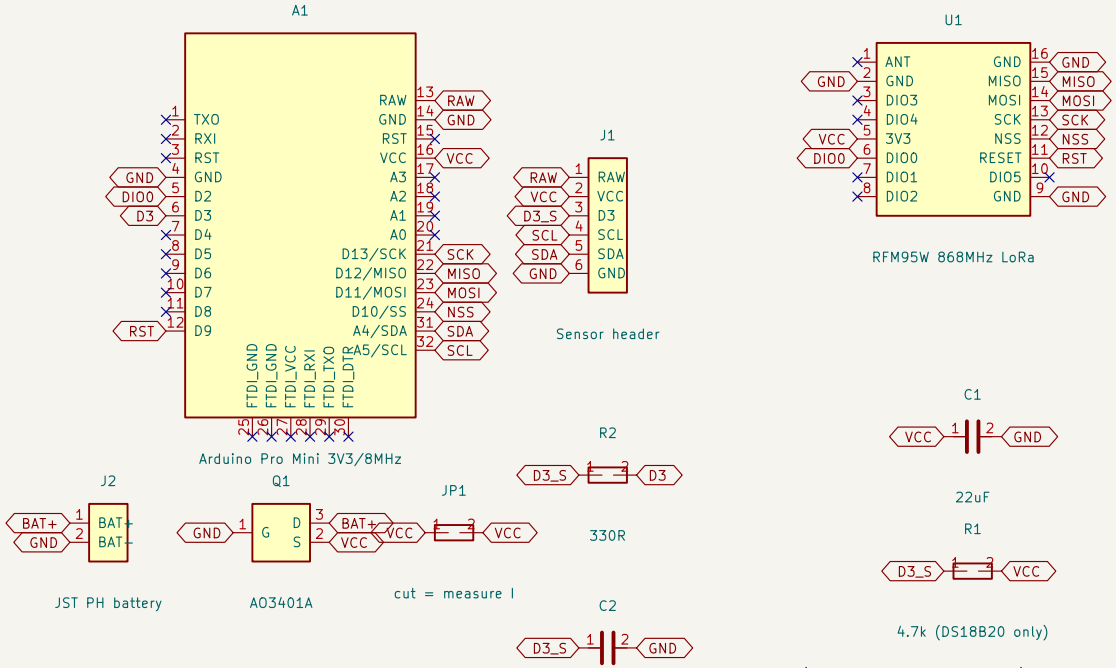


Carrier board: Arduino Pro Mini on top, RFM95W LoRa module on the bottom (muhradio LoRa node).

Use a 3.3 V / 8 MHz Pro Mini: the RFM95W is not 5 V tolerant (1.8–3.7 V).

Power: 1S LiPo / 18650 (or 2xAA) straight on VCC via J2 -> Q1 (reverse-polarity P-FET) -> JP1 (closed link, cut to insert an ammeter).

No regulator by user choice: a full Li-ion cell is 4.2 V, above the CC1101 3.6 V rating. Use protected cells (nothing here stops deep discharge).

Remove the Pro Mini power LED and regulator; unplug the battery while programming (FTDI adapter set to 3.3 V).

C1 22uF bulk for TX bursts. R1 4.7k = DS18B20 pull-up (DS18B20 nodes only). R2 330R series protection on D3.

C2 100nF = reed/button debounce, leave OFF for DS18B20 (1-Wire timing).

J1 sensor header: RAW VCC D3 SCL SDA GND. Radio: D13 SCK, D12 MISO, D11 MOSI, D10 NSS, D9 RST, D2 DIO0 (muhradio LoRa pinout).

A4/A5 holes match the SparkFun layout; on other clones jumper A4/A5 to J1 SDA/SCL.

FTDI row pads are plated but unconnected.

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Title: Arduino Pro Mini + RFM95W LoRa 868 MHz carrier

Size: A4	Date: 2026-09-14	**Rev: 1.3**
KiCad E.D.A. 10.0.6		Id: 1/1